

Hierarchical Models

1. 4.32
2. 4.35
3. For any two random variables X and Y with finite variances, prove that
 - (a) $E(XY) = E[XE(Y|X)]$
 - (b) $\text{Cov}(X, Y) = \text{Cov}(X, E(Y|X))$
 - (c) X and $Y - E(Y|X)$ are uncorrelated.
4. For each of the situations below, find $E(X)$.
 - (a) $X|Y \sim \text{Binomial}(n, Y)$ and $Y \sim \text{Beta}(a, b)$
 - (b) $X|Y \sim \text{Exponential}(Y)$ and $Y \sim \text{Gamma}(a, b)$ (*Use rate parametrization for both exponential and gamma*)
5. For part (a) of the previous problem, find the variance of X .
6. Suppose that $P(Y = 1) = .3$ and $P(Y = 2) = .7$ and that $X|Y \sim \text{Poisson}(Y)$. Find the mgf of X .

Covariance

7. CB 4.41
8. CB 4.42
9. CB 4.43
10. CB 4.65 (hint: Consider two i.i.d. random variables X, Y . Then, consider the expected value of the function $q = (g(x) - g(y))(h(x) - h(y))$. Consider $\mathbb{E}(q)$ by cases.)

Exponential Families

11. Show that each of the following families is an exponential family.
 - (a) Normal family with parameter μ known.
 - (b) Normal family with parameter σ known.
 - (c) Gamma family with parameter α known.
 - (d) Gamma family with parameter β known.
 - (e) Gamma family with both α and β unknown.
 - (f) Beta family with parameter α known.
 - (g) Beta family with parameter β known.
 - (h) Poisson family.
12. Show that each of the following families is an exponential family.
 - (a) Negative Binomial family with r known, $0 < p < 1$.

- (b) Pareto with α known.
 - (c) Beta family with both α and β unknown.
13. For each of the following families, write the pdf in the canonical (natural) parameterization of the exponential family (equation 3.4.7), and use the identities

$$E(t_j(X)) = -\frac{\partial}{\partial \eta_j} \log c^*(\eta) \qquad \text{Var}(t_j(X)) = -\frac{\partial^2}{\partial \eta_j^2} \log c^*(\eta)$$

to find the **mean** and the **variance** of X .

- (a) gamma(α, β)
 - (b) Poisson(λ)
 - (c) Binomial(n, p)
14. CB 3.33. For a and b, part (ii), you can use the parameters of the Normal parameters (mean and variance) or the parameterization given by the exponential family. For c and d, part (ii), use the parameterization given by the exponential family.